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Test Report issued under the responsibility of:

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TEST REPORT
IEC 62133-2:2017+AMD1:2021
Secondary cells and batteries containing alkaline or other non-acid electrolytes – Safety requirements for portable sealed secondary cells, and for batteries made from them, for use in portable applications – Part 2: Lithium systems

Report Number. :	RSZBHST2510112575
Date of issue :	2025-10-30
Total number of pages :	25
Name of Testing Laboratory preparing the Report :	Shenzhen Beihang Testing Co., Ltd.
Applicant's name :	HENZHEN YUNJI INTELLIGENT TECHNOLOGY CO.,LTD
Address :	202, Building A2, Silicon Valley Power Intelligent Terminal Industrial Park, No. 20, Dafu Industrial Zone, Kukeng Community, Guanlan Street, Longhua District, Shenzhen China
Manufacturer :	Zhongshanshi Jiuliyuan New Energy Technology Co.,Ltd.
Address :	Building 4, 2nd to 3rd floors, and Building 5, 2nd to 4th floors, No. 20 Beichen Road, Cuiheng New District, Nanlang Street, Zhongshan City
factory :	Zhongshanshi Jiuliyuan New Energy Technology Co.,Ltd.
Address :	Building 4, 2nd to 3rd floors, and Building 5, 2nd to 4th floors, No. 20 Beichen Road, Cuiheng New District, Nanlang Street, Zhongshan City
Model/Type reference :	S168
Trade Mark :	N/A
Ratings :	3.87V, 10000mAh,38.7Wh
Test item description :	Rechargeable Polymer Li-ion Battery
Standard :	IEC 62133-2:2017+AMD1:2021
Test procedure :	Test Report
Non-standard test method :	N/A

General disclaimer:

The test results presented in this report relate only to the object tested.

Responsible Testing Laboratory (as applicable), testing procedure and testing location(s):		
☒ **Testing Laboratory:**	Shenzhen Beihang Testing Co., Ltd.	
Testing location/ address.....:	Room 202， 2/F， Building F, HaoWei Industrial Park, QingSong West Road, PingShan District, Shenzhen, Guangdong, China/518000	
Tested by (name, function, signature).....:	Romy.Luo	
Approved by (name, function, signature)...:	Arvin.Shang	

Summary of testing:
The sample(s) tested complies with the requirements of IEC 62133-2:2017+AMD1:2021.

When determining the test conclusion, the Measurement Uncertainty of test has been considered.

Tests performed (name of test and test clause):	**Testing location:**
☒7.2.1 Continuous charging at constant voltage (cells) ☒7.3.1 External short circuit (cell) ☒7.3.2 External short circuit (battery) ☒7.3.3 Free fall ☒7.3.4 Thermal abuse (cells) ☒7.3.5 Crush (cells) ☒7.3.6 Over-charging of battery ☒7.3.7 Forced discharge (cells) ☒7.3.8 Mechanical tests (batteries) ☒7.3.9 Design evaluation – Forced internal short circuit (cells) ☐Annex D Measurement of the internal AC resistance for coin cells	Shenzhen Beihang Testing Co., Ltd. Room 202， 2/F， Building F, HaoWei Industrial Park, QingSong West Road, PingShan District, Shenzhen, Guangdong, China/518000

Summary of compliance with National Differences (List of countries addressed):

Group differences for CENELEC countries are considered.

The product fulfils the requirements of IEC 62133-2:2017+AMD1:2021 and EN 62133-2:2017+A1:2021

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Copy of marking plate:

The artwork below may be only a draft. The use of certification marks on a product must be authorized by the respective NCBs that own these marks.



Remark: “2025.04” represents the date of manufacture, “2025” represents year of manufacture, “04” represents the month of manufacture.

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IEC 62133-2			
Clause	Requirement + Test	Result - Remark	Verdict

Test item particulars:	--
Classification of installation and use:	Battery for special end products
Recommend charging method declared by the manufacturer:	Charging the battery with 2000mA constant current until 4.45V, then constant voltage until charge current reduces to 200mA at ambient 20°C±5°C
Discharge current (0.2 It A):	2000mA
Specified final voltage:	3.0V
Upper limit charging voltage per cell:	4.45V
Maximum charging current:	7000mA
Charging temperature upper limit:	60℃
Charging temperature lower limit:	0℃
Possible test case verdicts:	
- test case does not apply to the test object:	N/A
- test object does meet the requirement:	P (Pass)
- test object does not meet the requirement:	F (Fail)
Date of receipt of test item:	2025-10-11
Date (s) of performance of tests:	2025-10-12—2025-10-25

General remarks:

The test results presented in this report relate only to the object tested.
This report shall not be reproduced, except in full, without the written approval of the Issuing testing laboratory.
"(See Enclosure #)" refers to additional information appended to the report.
"(See appended table)" refers to a table appended to the report.

Throughout this report a ☐ comma / ☒ point is used as the decimal separator.

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Unless otherwise stated the results shown in this test report refer only to the sample(s) tested and such sample(s) are retained for 30 days only. This document cannot be reproduced except in full, without prior approval of the Company.

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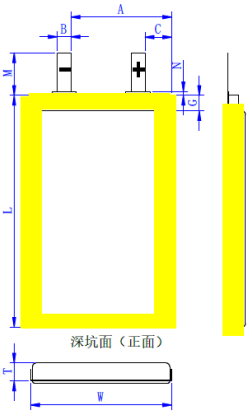
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IEC 62133-2			
Clause	Requirement + Test	Result - Remark	Verdict

General product information and other remarks:

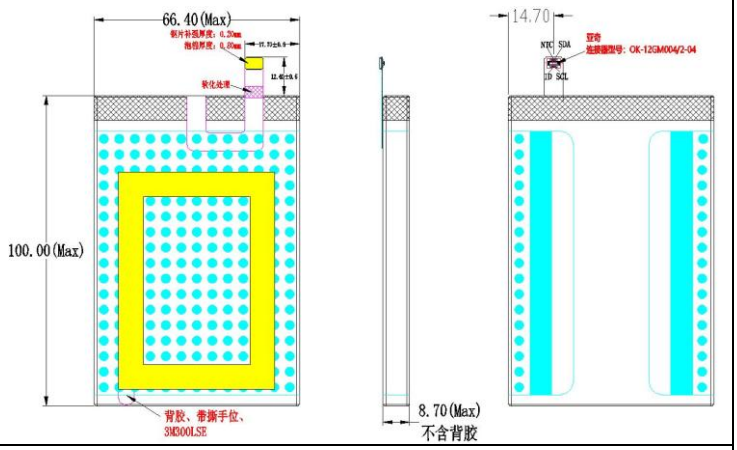
Product name	Rechargeable Polymer Li-ion Cell	Rechargeable Polymer Li-ion Battery
Type/model	426697	S168
Nominal voltage	3.87Vd.c.	3.87Vd.c.
Rated capacity	5000mAh	10000mAh
Recommended charging voltage	4.45V	4.45V
Nominal charging current	1000mAh	2000mAh
Nominal discharging current	1000mAh	2000mAh
Maximum charging current	5000mA	7000mA
Maximum discharging current	5000mA	7000mA
Discharge cut-off voltage	3.0V	3.0V
Charging temperature	0-60℃	0-60℃
charging cut off current	100mA	200mA

Dimension of the Cell：

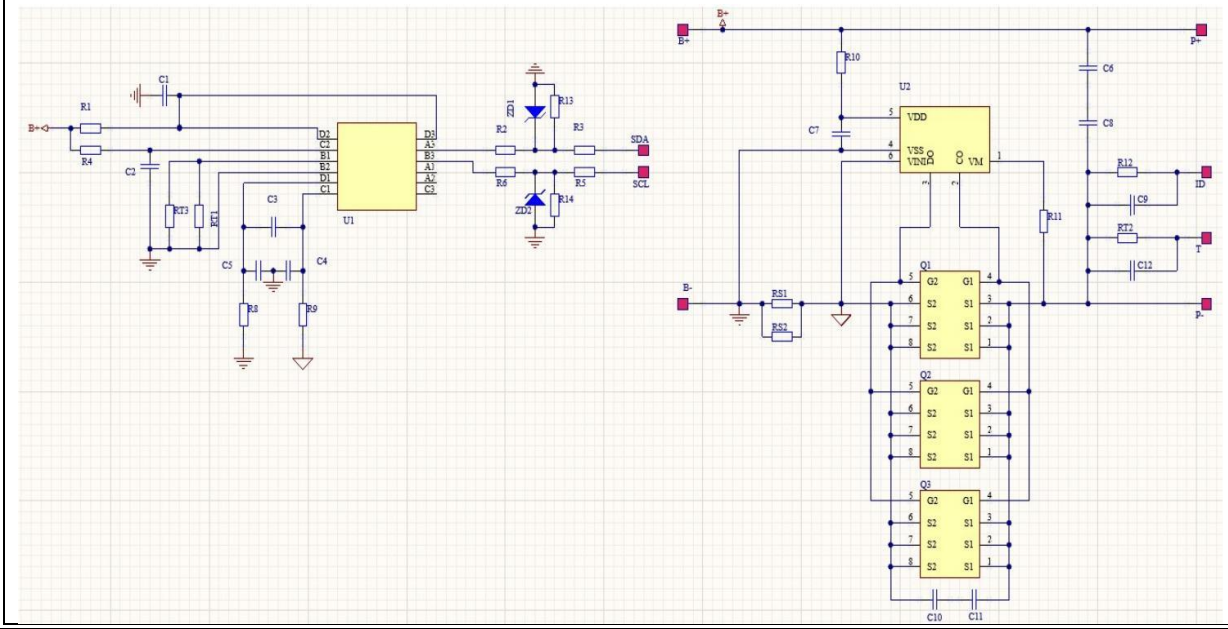


4.2mm(T) x 66.0mm(W) x 97.0mm(L)

Dimension of the Battery：



Protection Circuit：



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IEC 62133-2			
Clause	Requirement + Test	Result - Remark	Verdict
4	**PARAMETER MEASUREMENT TOLERANCES**		P
	Parameter measurement tolerances		P
5	**GENERAL SAFETY CONSIDERATIONS**		P
5.1	**General**		P
	Cells and batteries so designed and constructed that they are safe under conditions of both intended use and reasonably foreseeable misuse		P
5.2	**Insulation and wiring**		P
	The insulation resistance between the positive terminal and externally exposed metal surfaces of the battery (excluding electrical contact surfaces)is not less than 5 MΩ	No externally exposed metal surfaces.	N/A
	Insulation resistance (MΩ)..... :		—
	Internal wiring and insulation are sufficient to withstand maximum anticipated current, voltage and temperature requirements		P
	Orientation of wiring maintains adequate clearance and creepage distances between conductors		P
	Mechanical integrity of internal connections accommodates reasonably foreseeable misuse		P
5.3	**Venting**		P
	Battery cases and cells incorporate a pressure relief mechanism or are constructed so that they relieve excessive internal pressure at a value and rate that will preclude rupture, explosion and self-ignition	The vents on the narrow side of the cell	P
	Encapsulation used to support cells within an outer casing does not cause the battery to overheat during normal operation nor inhibit pressure relief	No encapsulation	N/A
5.4	**Temperature, voltage and current management**	See Below	P
	Batteries are designed such that abnormal temperature rise conditions are prevented	Overcharge, over discharge, over current and short-circuit proof circuit used in this battery. See tests of clause 8.	P
	Batteries are designed to be within temperature, voltage and current limits specified by the cell manufacturer	See above.	P
	Batteries are provided with specifications and charging instructions for equipment manufacturers so that specified chargers are designed to maintain charging within the temperature, voltage and current limits specified	The charging limits specified in the manufacturer's specifications.	P

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IEC 62133-2			
Clause	Requirement + Test	Result - Remark	Verdict
5.5	**Terminal contacts**		P
	The size and shape of the terminal contacts ensure that they can carry the maximum anticipated current	Complied.	P
	External terminal contact surfaces are formed from conductive materials with good mechanical strength and corrosion resistance	Complied.	P
	Terminal contacts are arranged to minimize the risk of short-circuit	Complied.	P
5.6	**Assembly of cells into batteries**		P
5.6.1	General		P
	Each battery have an independent control and protection for current, voltage, temperature and any other parameter required for safety and to maintain the cells within their operating region		P
	This protection may be provided external to the battery such as within the charger or the end devices		N/A
	If protection is external to the battery, the manufacturer of the battery provide this safety relevant information to the external device manufacturer for implementation		P
	If there is more than one battery housed in a single battery case, each battery have protective circuitry that can maintain the cells within their operating regions		N/A
	Manufacturers of cells specify current, voltage and temperature limits so that the battery manufacturer/designer may ensure proper design and assembly	Current, voltage and temperature limits specified by cell manufacturer.	P
	Batteries that are designed for the selective discharge of a portion of their series connected cells incorporate circuitry to prevent operation of cells outside the limits specified by the cell manufacturer		N/A
	Protective circuit components added as appropriate and consideration given to the end-device application		P
	The manufacturer of the battery provide a safety analysis of the battery safety circuitry with a test report including a fault analysis of the protection circuit under both charging and discharging conditions confirming the compliance		N/A
5.6.2	Design recommendation		P

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Clause	Requirement + Test	Result - Remark	Verdict
	For the battery consisting of a single cell or a single cellblock, it is recommended that the charging voltage of the cell does not exceed the upper limit of the charging voltage specified in Table 2	Single cell battery, charging voltage 4.45V, not exceed 4.45V specified in Table 2.	P
	For the battery consisting of series-connected plural single cells or series-connected plural cellblocks, it is recommended that the voltages of any one of the single cells or single cellblocks does not exceed the upper limit of the charging voltage, specified in Table 2, by monitoring the voltage of every single cell or the single cellblocks	1S2P	N/A
	For the battery consisting of series-connected plural single cells or series-connected plural cellblocks, it is recommended that charging is stopped when the upper limit of the charging voltage is exceeded for any one of the single cells or single cellblocks by measuring the voltage of every single cell or the single cellblocks	1S2P	N/A
	For batteries consisting of series-connected cells or cell blocks, nominal charge voltage not be counted as an overcharge protection	1S2P	N/A
	For batteries consisting of series-connected cells or cell blocks, cells have closely matched capacities, be of the same design, be of the same chemistry and be from the same manufacturer	1S2P	N/A
	It is recommended that the cells and cell blocks not discharged beyond the cell manufacturer's specified final voltage	Battery specified final voltage 3.0V, not exceed 3.0V specified by cell manufacturer	P
	For batteries consisting of series-connected cells or cell blocks, cell balancing circuitry incorporated into the battery management system	1S2P	N/A
5.6.3	Mechanical protection for cells and components of batteries		P
	Mechanical protection for cells, cell connections and control circuits within the battery provided to prevent damage as a result of intended use and reasonably foreseeable misuse	The battery has a gummed case	P
	The mechanical protection can be provided by the battery case or it can be provided by the end product enclosure for those batteries intended for building into an end product	Build-in batteries , The mechanical protection are provided by the end product enclosure	N/A
	The battery case and compartments housing cells designed to accommodate cell dimensional tolerances during charging and discharging as recommended by the cell manufacturer	No battery case	N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	For batteries intended for building into a portable end product, testing with the battery installed within the end product considered when conducting mechanical tests		P
5.7	**Quality plan**		P
	The manufacturer prepares and implements a quality plan that defines procedures for the inspection of materials, components, cells and batteries and which covers the whole process of producing each type of cell or battery	Complied. ISO 9001: 2015 certificate provided.	P
5.8	**Battery safety components**		N/A
	According annex F	See TABLE: Critical components information.	N/A
6	**TYPE TEST AND SAMPLE SIZE**		P
	Tests are made with the number of cells or batteries specified in Table 1 using cells or batteries that are not more than six months old	Tests are performed according to specified in Table 1 of this standard. The samples are not more than six months old	P
	Coin cells with resistance ≤ 3 Ω (measured according annex D) are tested according table 1	Not Coin cell	N/A
	Unless otherwise specified, tests are carried out in an ambient temperature of 20 °C ± 5 °C	Tests are carried out at 20°C ± 5°C.	P
	The safety analysis of 5.6.1 identify those components of the protection circuit that are critical for short-circuit, overcharge and over discharge protection		P
	When conducting the short-circuit test, consideration given to the simulation of any single fault condition that is likely to occur in the protecting circuit that would affect the short-circuit test	See 7.3.2	P
7	**SPECIFIC REQUIREMENTS AND TESTS**		P
7.1	**Charging procedure for test purposes**		P
7.1.1	First procedure		P
	This charging procedure applies to subclauses other than those specified in 7.1.2		P
	Unless otherwise stated in this document, the charging procedure for test purposes is carried out in an ambient temperature of 20 °C ± 5 °C, using the method declared by the manufacturer		P
	Prior to charging, the battery have been discharged at 20 °C ± 5 °C at a constant current of 0,2 It A down to a specified final voltage		P

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IEC 62133-2			
Clause	Requirement + Test	Result - Remark	Verdict
7.1.2	Second procedure		P
	This charging procedure applies only to 7.3.1, 7.3.4, 7.3.5, and 7.3.9		P
	After stabilization for 1 h to 4 h at an ambient temperature of the highest test temperature and the lowest test temperature, respectively, as specified in Table 2, cells are charged by using the upper limit charging voltage and maximum charging current, until the charging current is reduced to 0,05 I t A, using a constant current to constant voltage charging method.	Charging temperature specified by client is 0-60 ℃，60 ℃ and 0℃ were used as highest test temperature and lowest test temperature during tests.	P
7.2	**Intended use**		P
7.2.1	Continuous charging at constant voltage (cells)		P
	Fully charged cells are subjected for 7 days to a charge using the charging method for current and standard voltage specified by the cell manufacturer	4.45V/1000mA	P
	Results: No fire. No explosion. No leakage.....:	(See appended table 7.2.1)	P
7.2.2	Case stress at high ambient temperature (battery)	Tested complied.	P
	Oven temperature (°C).....:	70	—
	Results :No physical distortion of the battery case resulting in exposure of internal protective components and cells	No physical distortion of the battery case resulting in exposure of internal protective components and cells.	P
7.3	**Reasonably foreseeable misuse**		P
7.3.1	External short-circuit (cell)	Tested complied.	P
	The cells were tested until one of the following occurred:		P
	- 24 hours elapsed; or		N/A
	- The case temperature declined by 20% of the maximum temperature rise		P
	Results: No fire. No explosion.....:	(See appended table7.3.1)	P
7.3.2	External short-circuit (battery)	Tested complied.	P
	The batteries were tested until one of the following occurred:		P
	- 24 hours elapsed; or		N/A
	- The case temperature declined by 20 % of the maximum temperature rise		P
	In case of rapid decline in short circuit current, the battery pack remained on test for an additional one hour after the current reached a low end steady state condition		P

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IEC 62133-2			
Clause	Requirement + Test	Result - Remark	Verdict
	A single fault in the discharge protection circuit conducted on one to four (depending upon the protection circuit) of the five samples before conducting the short-circuit test	Single fault conducted on 3 samples.	P
	A single fault applies to protective component parts such as MOSFET, fuse, thermostat or positive temperature coefficient (PTC) thermistor	Single fault applies on MOSFET. Q1（pin3-pin6）	P
	Results: No fire. No explosion..... :	(See appended table7.3.2)	P
7.3.3	Free fall	1m	P
	Results: No fire. No explosion	No fire. No explosion	P
7.3.4	Thermal abuse (cells)	Tested complied.	P
	Oven temperature (°C)..... :	130℃,30min	—
	Results: No fire. No explosion	No fire. No explosion	P
7.3.5	Crush (cells)	Tested complied.	P
	The crushing force was released upon:		P
	- The maximum force of 13 kN±0,78kN has been applied; or	13 kN	P
	- An abrupt voltage drop of one-third of the original voltage has been obtained		N/A
	Results: No fire. No explosion..... :	(See appended table 7.3.5)	P
7.3.6	Over-charging of battery		P
	The supply voltage which is:		P
	- 1,4 times the upper limit charging voltage presented in Table A.1 (but not to exceed 6,0 V) for single cell/cell block batteries or	6.0V	P
	- 1,2 times the upper limit charging voltage resented in Table A.1 per cell for series connected multi-cell batteries, and		N/A
	- Sufficient to maintain a current of 2,0 It A throughout the duration of the test or until the supply voltage is reached	20.0A	P
	Test was continued until the temperature of the outer casing:		P
	- Reached steady state conditions (less than 10°C change in 30-minute period); or		N/A
	- Returned to ambient		P
	Results: No fire. No explosion..... :	(See appended table7.3.6)	P
7.3.7	Forced discharge (cells)		P

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Clause	Requirement + Test	Result - Remark	Verdict
	If the discharge voltage reaches the negative value of upper limit charging voltage within the testing duration, the voltage is maintained at the negative value of the upper limit charging voltage by reducing the current for the remainder of the testing duration		N/A
	If the discharge voltage does not reach the negative value of upper limit charging voltage within the testing duration, the test is terminated at the end of the testing duration		P
	Results: No fire. No explosion..... :	(See appended table7.3.7)	P
7.3.8	Mechanical tests (batteries)		P
7.3.8.1	Vibration		P
	Results: No fire, no explosion, no rupture, no leakage or venting. ..... :	Tested complied.	P
7.3.8.2	Mechanical shock		P
	Results: No leakage, no venting, no rupture, no explosion and no fire ..... :		P
7.3.9	Design evaluation – Forced internal short-circuit (cells)	Tested complied.	P
	The cells complied with national requirement for .....:	France, Japan, Korea, Switzerland	—
	The pressing was stopped upon:		P
	- A voltage drop of 50 mV has been detected; or		N/A
	- The pressing force of 800 N (cylindrical cells) or 400 N (prismatic cells) has been reached	400 N (prismatic cells)	P
	Results: No fire ..... :	(See appended table7.3.9)	P
8	**INFORMATION FOR SAFETY**		P
8.1	**General**		P
	Manufacturers of secondary cells ensure that information is provided about current, voltage and temperature limits of their products	Information for safety mentioned in manufacturer's specifications.	P
	Manufacturers of batteries ensure that equipment manufacturers and, in the case of direct sales, end-users are provided with information to minimize and mitigate hazards	Information for safety mentioned in manufacturer's specifications.	P
	Systems analyses performed by device manufacturers to ensure that a particular battery design prevents hazards from occurring during use of a product		P

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Clause	Requirement + Test	Result - Remark	Verdict
	As appropriate, any information relating to hazard avoidance resulting from a system analysis provided to the end user		P
	Do not allow children to replace batteries without adult supervision	Big enough	N/A
8.2	**Small cell and battery safety information**	Big enough	N/A
	The following warning language is to be provided with the information packaged with the small cells and batteries or equipment using them:		N/A
	- Keep small cells and batteries which are considered swallowable out of the reach of children		N/A
	- Swallowing may lead to burns, perforation of soft tissue, and death. Severe burns can occur within 2 h of ingestion		N/A
	- In case of ingestion of a cell or battery, seek medical assistance promptly		N/A
9	**MARKING**		P
9.1	**Cell marking**	The final product is battery.	N/A
	Cells marked as specified in IEC 61960, except coin cells		N/A
	Coin cells whose external surface area is too small to accommodate the markings on the cells show the designation and polarity		N/A
	By agreement between the cell manufacturer and the battery and/or end product manufacturer, component cells used in the manufacture of a battery need not be marked		N/A
9.2	**Battery marking**		P
	Batteries marked as specified in IEC 61960, except for coin batteries	The battery is marked in accordance with IEC 61960,	P
	Coin batteries whose external surface area is too small to accommodate the markings on the batteries show the designation and polarity. Batteries also marked with an appropriate caution statement	Not coin battery.	N/A
	Terminals have clear polarity marking on the external surface of the battery	see page 4.	P
	Batteries with keyed external connectors designed for connection to specific end products need not be marked with polarity markings if the design of the external connector prevents reverse polarity connections	FPCB	P

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Clause	Requirement + Test	Result - Remark	Verdict
9.3	**Caution for ingestion of small cells and batteries**		N/A
	Coin cells and batteries identified as small batteries according to 8.2 include a caution statement regarding the hazards of ingestion in accordance with 8.2	Not coin cell or battery.	N/A
	When small cells and batteries are intended for direct sale in consumer-replaceable applications, caution for ingestion given on the immediate package	Not intended for direct sale in consumer-replaceable applications.	N/A
9.4	**Other information**		P
	Storage and disposal instructions	Information for disposal instructions mentioned in manufacturer's specifications.	P
	Recommended charging instructions	Information for disposal instructions mentioned in manufacturer's specifications.	P
10	**PACKAGING AND TRANSPORT**		P
	Packaging for coin cells not small enough to fit within the limits of the ingestion gauge of Figure 3	Not coin cells	N/A
	The materials and packaging design are chosen so as to prevent the development of unintentional electrical conduction, corrosion of the terminals and ingress of environmental contaminants		P
ANNEX A	**CHARGING AND DISCHARGING RANGE OF SECONDARY LITHIUM ION CELLS FOR SAFE USE**		P
A.1	**General**		P
A.2	**Safety of lithium ion secondary battery**	Complied.	P
A.3	**Consideration on charging voltage**	Complied.	P
A.3.1	General		P
A.3.2	Upper limit charging voltage	4.45V applied.	P
A.3.2.1	General		P
A.3.2.2	Explanation of safety viewpoint		N/A
A.3.2.3	Safety requirements, when different upper limit charging voltage is applied	4.45V applied.	N/A
A.4	**Consideration of temperature and charging current**		P
A.4.1	General		P
A.4.2	Recommended temperature range	See A.4.2.2.	P
A.4.2.1	General		P

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Clause	Requirement + Test	Result - Remark	Verdict
A.4.2.2	Safety consideration when a different recommended temperature range is applied	Charging temperature range declared by client is: 0-60°C	P
A.4.3	High temperature range	Charging high temperature declared by client is: 60°C	N/A
A.4.3.1	General		N/A
A.4.3.2	Explanation of safety viewpoint		N/A
A.4.3.3	Safety considerations when specifying charging conditions in the high temperature range		N/A
A.4.3.4	Safety considerations when specifying a new upper limit in the high temperature range	60°C applied.	N/A
A.4.4	Low temperature range	Charging low temperature declared by client is: 0°C	N/A
A.4.4.1	General		N/A
A.4.4.2	Explanation of safety viewpoint		N/A
A.4.4.3	Safety considerations, when specifying charging conditions in the low temperature range		N/A
A.4.4.4	Safety considerations when specifying a new lower limit in the low temperature range	0°C applied.	N/A
A.4.5	Scope of the application of charging current		P
A.4.6	Consideration of discharge		P
A.4.6.1	General		P
A.4.6.2	Final discharge voltage and explanation of safety viewpoint		P
A.4.6.3	Discharge current and temperature range		P
A.4.6.4	Scope of application of the discharging current		P
A.5	**Sample preparation**		P
A.5.1	General		P
A.5.2	Insertion procedure for nickel particle to generate internal short		P
A.5.3	Disassembly of charged cell		P
A.5.4	Shape of nickel particle		P
A.5.5	Insertion of nickel particle in cylindrical cell		N/A
A.5.5.1	Insertion of nickel particle in winding core		P
A.5.5.2	Marking the position of the nickel particle on both ends of the winding core of the separator		P
A.5.6	Insertion of nickel particle in prismatic cell		P
A.6	**Experimental procedure of the forced internal short-circuit test**		P

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Clause	Requirement + Test	Result - Remark	Verdict
A.6.1	Material and tools for preparation of nickel particle		P
A.6.2	Example of a nickel particle preparation procedure		P
A.6.3	Positioning (or placement) of a nickel particle		P
A.6.4	Damaged separator precaution		P
A.6.5	Caution for rewinding separator and electrode		P
A.6.6	Insulation film for preventing short-circuit		P
A.6.7	Caution when disassembling a cell		P
A.6.8	Protective equipment for safety		P
A.6.9	Caution in the case of fire during disassembling		P
A.6.10	Caution for the disassembling process and pressing the electrode core		P
A.6.11	Recommended specifications for the pressing device		P
ANNEX B	**RECOMMENDATIONS TO EQUIPMENT MANUFACTURERS AND BATTERY ASSEMBLERS**		P
ANNEX C	**RECOMMENDATIONS TO THE END-USERS**		P
ANNEX D	**MEASUREMENT OF THE INTERNAL AC RESISTANCE FOR COIN CELLS**		N/A
D.1	**General**	Not coin cells	N/A
D.2	**Method**		N/A
	A sample size of three coin cells is required for this measurement..... :	(See appended tableD.2)	N/A
	Coin cells with an internal resistance of less than or equal to 3 Ω are subjected to the testing according to Clause 6 and Table 1		N/A
	Coin cells with an internal resistance greater than 3 Ω require no further testing		N/A
ANNEX E	**PACKAGING AND TRANSPORT**		P
ANNEX F	**COMPONENT STANDARDS REFERENCES**		N/A

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Clause	Requirement + Test	Result - Remark	Verdict

	TABLE: Critical components information				
Object / part No.	**Manufacturer/trademark**	**Type / model**	**Technical data**	**Standard**	**Mark(s) of conformity[1)]**
Cell	Shenzhenshi Jiuliyuan Electronic Technology Co.,Ltd	426697	3.87V,5000mAh	--	Tested with appliance
FPCB	JRD/LTD	RR-JLY-S168-426697-4.45V-PCB	62.3*7.5* 1.0mm	--	Tested with appliance
FPC	FLD/TLD	RR-JLY-S168-426697-4.45V-A1-FPC	30.7*27.09*0.2mm	--	Tested with appliance
Protection IC (U1)	CMW	CM1004-CMD	Overcharge detection voltage: 4.475V±0.025V, Overdischarge detection voltage: 2.5V±0.05V,	--	Tested with appliance
MOSFET (Q1、Q2、Q3)	ITM	TMM1817E	VDS: 20V, VGS: ±12V, ID: 5A(TA=25°C), TJ: -55°C to 150°C	--	Tested with appliance
NTC	CUNTIAN	NCP03XH103F05RL	10KΩ,±1%,B=3380	--	--
Supplementary information: 1) Provided evidence ensures the agreed level of compliance. See OD-CB2039.					

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Clause	Requirement + Test	Result - Remark	Verdict

7.2.1	TABLE: Continuous charging at constant voltage (cells)			P
Sample no.	**Recommended charging voltage Vc (Vdc)**	**Recommended charging current I_{rec} (A)**	**OCV before test(Vdc)**	**Results**
C1	4.45	2.0	4.377	P
C2	4.45	2.0	4.373	P
C3	4.45	2.0	4.372	P
C4	4.45	2.0	4.379	P
C5	4.45	2.0	4.376	P

Supplementary information:

- No fire or explosion
- No leakage
- Others (please explain)

7.3.1	TABLE: External short-circuit (cell)				P
Sample no.	**Ambient T (°C)**	**OCV before test (Vdc)**	**Resistance of circuit (mΩ)**	**Maximum case temperature (℃)**	**Results**
Samples charged at charging temperature upper limit（60℃）					
C6	55.3	4.411	80	118.5	P
C7	55.3	4.409	80	121.3	P
C8	55.3	4.412	80	120.7	P
C9	55.3	4.404	80	119.3	P
C10	55.3	4.406	80	117.4	P
Samples charged at charging temperature lower limit（0℃）					
C11	55.3	4.286	80	121.5	P
C12	55.3	4.285	80	122.3	P
C13	55.3	4.288	80	117.8	P
C14	55.3	4.284	80	120.6	P
C15	55.3	4.287	80	118.4	P

Supplementary information:

- No fire or explosion
- Others (please explain)

7.3.2	TABLE: External short-circuit (battery)	P

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Clause	Requirement + Test	Result - Remark	Verdict

Sample no.	Ambient T (°C)	OCV before test (Vdc)	Resistance of circuit (mΩ)	Maximum case temperature (℃)	Component single fault condition	Results
B1	23.5	4.378	80	120.7	MOSFET（Q1）	P
B2	23.5	4.373	80	118.9	MOSFET（Q1）	P
B3	23.5	4.375	80	121.3	MOSFET（Q1）	P
B4	23.5	4.377	80	24.4	Normal	P
B5	23.5	4.374	80	24.2	Normal	P

Supplementary information:

- No fire or explosion
- Others (please explain)

Mosfet (pinS-C) short circuit.

7.3.5	TABLE: Crush (cells)			P
Sample no.	**OCV before test (Vdc)**	**OCV at removal of crushing force (Vdc)**	**Maximum force applied to the cell during crush (kN)**	**Results**
Samples charged at charging temperature upper limit（60℃）				
C16	4.408	4.406	13.0	P
C17	4.411	4.411	13.0	P
C18	4.405	4.404	13.0	P
C19	4.407	4.406	13.0	P
C20	4.412	4.411	13.0	P
Samples charged at charging temperature lower limit（0℃）				
C21	4.286	4.285	13.0	P
C22	4.288	4.286	13.0	P
C23	4.284	4.283	13.0	P
C24	4.286	4.286	13.0	P
C25	4.287	4.285	13.0	P

Supplementary information:

- No fire or explosion
- Others (please explain)

7.3.6	TABLE: Over-charging of battery	P
Constant charging current (A) .....:	**20.0A**	—
Supply voltage (Vdc) .....:	**6.0V**	—

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Clause	Requirement + Test	Result - Remark	Verdict

Sample no.	OCV before charging (Vdc)	cut-off condition	Maximum outer case temperature (°C)	Results
B6	3.343	90min	34.2	P
B7	3.337	90min	34.7	P
B8	3.342	90min	32.9	P
B9	3.339	90min	33.8	P
B10	3.343	90min	33.4	P

Supplementary information:

- No fire or explosion
- Others (please explain)

7.3.7	TABLE: Forced discharge (cells)	P

Sample no.	OCV before application of reverse charge (Vdc)	Measured reverse charge I_t (A)	Lower limit discharge voltage (Vdc)	Results
C26	3.337	5.0	3.0	P
C27	3.342	5.0	3.0	P
C28	3.339	5.0	3.0	P
C29	3.343	5.0	3.0	P
C30	3.344	5.0	3.0	P

Supplementary information:

- No fire or explosion
- Others (please explain)

7.3.8.1	TABLE: Vibration	P

Sample no.	OCV before test (Vdc)	OCV after test (Vdc)	Mass before test (g)	Mass after test(g)	Results
B11	4.384	4.382	132.294	132.292	P
B12	4.382	4.381	132.139	132.136	P
B13	4.386	4.384	132.178	132.175	P

Supplementary information:

- No fire or explosion
- No rupture
- No leakage
- No venting
- Others (please explain)

7.3.8.2	TABLE: Mechanical shock	P

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Clause	Requirement + Test	Result - Remark	Verdict

Sample no.	OCV before test (Vdc)	OCV after test (Vdc)	Mass before test (g)	Mass after test(g)	Results
B14	4.384	4.384	132.237	132.237	P
B15	4.388	4.388	132.206	132.206	P
B16	4.387	4.387	132.296	132.296	P

Supplementary information:

- No fire or explosion
- No rupture
- No leakage
- No venting
- Others (please explain)

7.3.9	TABLE: Forced internal short circuit (cells)				P
Sample no.	**Chamber ambient T (°C)**	**OCV before test (Vdc)**	**Particle location[1)]**	**Maximum applied pressure (N)**	**Results**
Samples charged at charging temperature upper limit（60℃）					
C31	60	4.389	1	400	P
C32	60	4.383	1	400	P
C33	60	4.387	1	400	P
C34	60	4.388	2	400	P
C35	60	4.384	2	400	P
Samples charged at charging temperature lower limit（0℃）					
C36	0	4.287	1	400	P
C37	0	4.285	1	400	P
C38	0	4.283	1	400	P
C39	0	4.284	2	400	P
C40	0	4.282	2	400	P

Supplementary information:

[1)]Identify one of the following:

1: Nickel particle inserted between positive and negative (active material) coated area.

2: Nickel particle inserted between positive aluminium foil and negative active material coated area.

- No fire or explosion
- Others (please explain)

D.2	TABLE: Internal AC resistance for coin cells			N/A
Sample no.	**Ambient T (°C)**	**Store time (h)**	**Resistance Rac (Ω)**	**Results[1)]**

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Clause	Requirement + Test	Result - Remark	Verdict

Supplementary information:

1) Coin cells with internal resistance less than or equal to 3 Ω, see test result on corresponding tables

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Photos

Fig.1 –Front view of battery



Fig. 2 –Back view of battery



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Fig.3 –Front view of Cell



Fig.4 –Back view of Cell



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Fig.5 –Front view of PCM

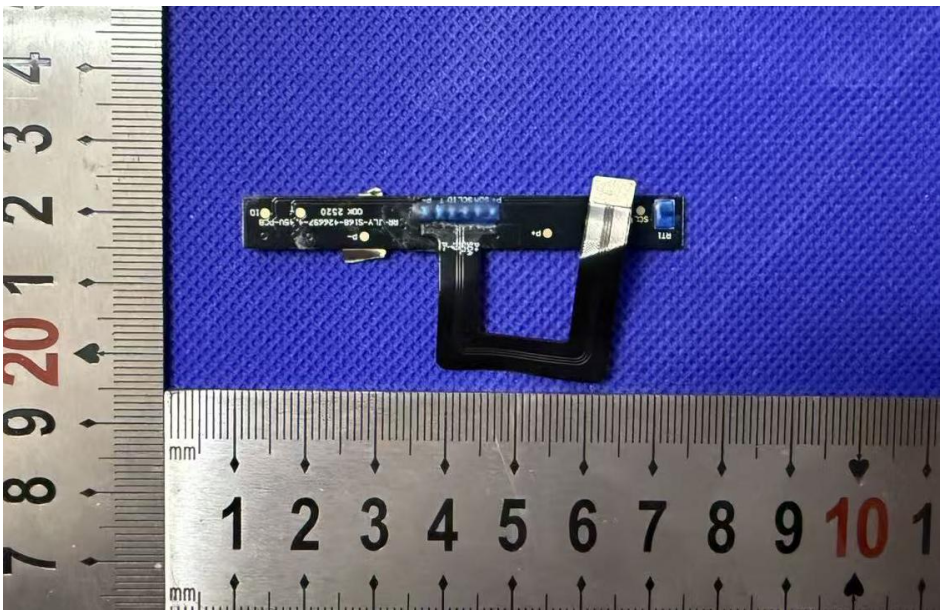
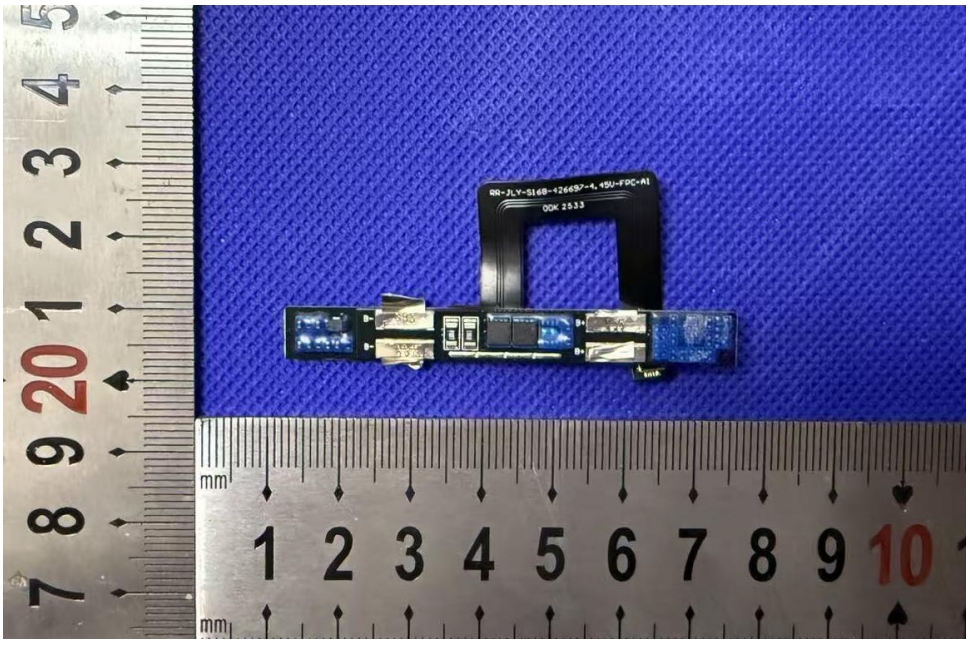


Fig.6 –Back view of PCM



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